

Transverse O⁺ heating does not explain sub-threshold auroral ion-cyclotron waves — it suppresses them

Claim. Electrostatic ion-cyclotron (EIC) waves are seen in the topside auroral ionosphere at field-aligned currents below the Kindel & Kennel (1971) threshold. Transversely heated O⁺ is co-observed with them and carries perpendicular free energy the current-based threshold ignores, so it is a natural candidate to close the gap. It does not. A **minority** heated population — the observed situation in the topside — raises the current threshold, by 30% at a ring fraction $\delta = 0.3$ and by 58% when the whole O⁺ is a ring too broad to be unstable on its own. The currentless channel exists but is demanding: it needs a sharp ring ($v_r/w \gtrsim 1.44$, $E_\perp \gtrsim 0.63$ eV) that also **dominates** the O⁺ population ($\delta \gtrsim 0.6$ – 0.9). The two free-energy sources compete rather than add. Separately, and robustly: at identical $(n, T_\perp, T_\parallel)$ a ring and a bi-Maxwellian sit at opposite extremes — no current required versus four times the isotropic threshold — so this threshold is not a function of the moments that observational studies report.

1 Why this question

Auroral field-aligned currents flow through a topside ionosphere that is cold ($T \sim 0.3$ eV), dense, and dominated by O⁺. Kindel & Kennel (JGR 76, 3055, 1971) showed that such a current destabilises electrostatic ion-cyclotron waves before anything else, and that result has framed auroral wave physics since: EIC waves are read as a current diagnostic.

The reading has a long-standing problem. EIC waves are observed when the measured field-aligned drift sits **below** the critical value. The accepted resolution (Gavrishchaka et al. 2000; Ganguli et al. 2002; Peñano & Ganguli, JGR 107, 1315, 2002) adds a second free-energy source: shear in the ion flow changes the sign of the cyclotron damping term, so EIC waves grow with no current at all.

But the same altitudes host **transverse ion heating**. Ion conics and transversely accelerated ions (TAI) are among the best-documented features of the auroral topside (Retterer et al., JGR 99, 13189, 1994). A transversely heated population is not a displaced Maxwellian: it is a ring or shell in $v_\perp$ with $\partial f_0 / \partial v_\perp > 0$ on its inner flank — free energy of exactly the kind cyclotron harmonics feed on. That this works in principle is established elsewhere in geospace: oxygen cyclotron harmonic waves seen by Akebono, Van Allen Probes and MMS are attributed to the O⁺ ion Bernstein instability driven by ring-like O⁺ (Min et al., JGR 122, 2017 Liu et al., GRL 47, 2020 and JGR 127, 2022), and pickup-ion rings drive the same family at Mars.

So the question is narrow: **does the transversely heated O⁺ already present in the auroral topside destabilise the ion-cyclotron band by itself, and does it lower the current the band would otherwise need?** If it does, the sub-threshold observations need no shear.

The reason this has not been settled is decisive: the free energy lives in the shape of f_0 , and the dispersion solvers in routine use accept only bi-Maxwellian or kappa distributions — a ring cannot be written in their inputs. Answering it needs a hot magnetized kinetic solver taking an arbitrary gyrotropic f_0 ; here VlasovMaxwellDispersion.jl (VMD).

2 Setup

Reference plasma. Topside auroral, $B_0 = 2.8 \times 10^{-5}$ T, $n = 3 \times 10^3$ cm⁻³, O⁺ with $T_i = 0.3$ eV and $T_e/T_i = 1$ unless stated. Then $\Omega_{O^+}/2\pi = 26.7$ Hz, $\rho_{O^+} = 11.3$ m, $w_i = 1.9$ km/s, $(\omega_{pi}/\Omega_i)^2 = 1.16 \times 10^4$. Ion-neutral collisions are negligible above ~ 250 km ($\nu_{in}/\Omega_i \lesssim 10^{-2}$).

Model. An O⁺ core (fraction $1 - \delta$, isotropic Maxwellian), an O⁺ ring (fraction δ , gyro-ring I_0 Maxwellian at ring speed v_r and the same thermal width w), and Maxwellian electrons drifting at u along $\mathbf{B}$. The knobs are the **ring sharpness** v_r/w — equivalently $E_\perp = T_i(v_r/w)^2$ — the **ring fraction** δ , and the drift u , reported in electron thermal speeds and as $j = neuv_e$.

Solve. Electrostatic reduction (mode = :P), which is the Kindel & Kennel dispersion relation and is accurate at these $\beta \sim 10^{-4}$. Each parameter point runs seedless box surveys over $k_{\perp}\rho \in [0.5, 3.5]$, $k_{\parallel}/k_{\perp} \in [10^{-4}, 0.12]$, taking the peak growth; thresholds are bisected to $\gamma = 10^{-3}\Omega_{O^+}$. Every growing root in §4 was re-solved with a seeded Muller polish and returned to the same place.

3 Gate: the current-driven threshold

The survey finds the unstable root just above the first gyroharmonic at $k_{\perp}\rho_i \approx 1.5$ — textbook EIC — with a threshold falling as T_e/T_i rises and ion Landau damping weakens:

T_e/T_i	u_c/v_e	u_c/w_i	j_c [$\mu\text{A}/\text{m}^2$]	$k_{\perp}\rho$	$k_{\parallel}/k_{\perp}$	ω_r/Ω
0.5	0.208	25.2	23.0	1.50	0.035	1.111
1.0	0.106	18.1	16.5	1.50	0.050	1.186
2.0	0.059	14.2	12.9	1.25	0.080	1.293
4.0	0.034	11.7	10.7	1.25	0.120	1.433

Table 1: Current-driven EIC threshold, pure Maxwellian O^+ . $u_c/w_i \approx 12$ –25 reproduces Kindel & Kennel.

The threshold drift is a property of the temperature ratio, not the density: $u_c/v_e = 0.1055$ to four digits across a hundredfold change in n , since $k\lambda_{De} \sim 10^{-2}$ leaves the Debye term negligible. Density enters the **current** only through $j = neu_c v_e$.

This is the quantitative form of the puzzle: the classical threshold is $16.5 \mu\text{A}/\text{m}^2$ here, while auroral field-aligned currents are typically 1 – $10 \mu\text{A}/\text{m}^2$. The waves are observed anyway.

4 The ring on its own does work — if it is sharp and it dominates

Set $u = 0$: no current anywhere. Sharpening the ring destabilises the band at

$$v_r/w = 1.44, \text{ i.e. } E_{\perp} = 0.63 \text{ eV, at } \omega_r = 0.99\Omega_{O^+}$$

and growth then rises steeply — $\gamma = 0.04\Omega$ by $E_{\perp} = 1.2 \text{ eV}$ and 0.19Ω by 1.9 eV , e-folding within a few gyroperiods. The unstable frequency hops between gyroharmonics as the ring sharpens ($\omega_r/\Omega = 0.99 \rightarrow 2.00 \rightarrow 0.51 \rightarrow 4.06$), the multi-band signature characteristic of ring-driven Bernstein modes.

That threshold energy is undemanding — well below observed conic energies. The binding constraint is elsewhere. Requiring instability at fixed v_r and solving instead for the **fraction** of O^+ that must be ring-like:

v_r/w	$E_{\perp}$ [eV]	δ_c
2.0	1.2	0.91
2.5	1.9	0.82
3.0	2.7	0.59

Table 2: Ring fraction needed for currentless instability. The heated population must be most of the O^+ , not a suprathermal minority.

5 The mechanisms compete, they do not add

The observationally relevant configuration is a heated minority over a dense cold core. There, transverse heating moves the current threshold the **wrong way**:

$E_{\perp}$ [eV]	$j_c, \delta = 0.1$	$j_c, \delta = 0.3$	$j_c, \delta = 1$
0.00	16.5	16.5	16.5
0.075	16.6	16.9	18.9
0.30	16.9	18.9	26.1
0.67	17.4	20.4	0
1.20	17.5	21.5	0
1.88	17.4	20.1	0

Table 3: Threshold current [$\mu\text{A}/\text{m}^2$] versus how heated the O^+ already is. A minority ring **raises** it; a dominant ring raises it further still – until it crosses its own threshold, where the requirement vanishes.

The $\delta = 1$ column is the sharpest statement. Below its own currentless threshold a fully ring-like O^+ makes the current-driven mode **harder** to excite by 58% ($16.5 \rightarrow 26.1 \mu\text{A}/\text{m}^2$ at $E_{\perp} = 0.30 \text{ eV}$); just past it, the requirement drops discontinuously to zero. There is no regime in which transverse heating gently assists the current.

Physically this is not surprising once seen. The current-driven mode at $\omega \approx 1.19\Omega$, $k_{\perp}\rho \approx 1.5$ survives because the ion perpendicular distribution is narrow enough to keep cyclotron damping small. A hot ring component broadens $\langle v_{\perp}^2 \rangle$, pushing the ring ions to larger $k_{\perp}\rho$ and **increasing** the damping of precisely that mode, while the ring’s own unstable modes live at different (k, ω) and need the ring to dominate before they overcome the cold core’s damping. The two channels never share a mode: they compete for the same population, and below the ring’s own threshold the heating is pure damping.

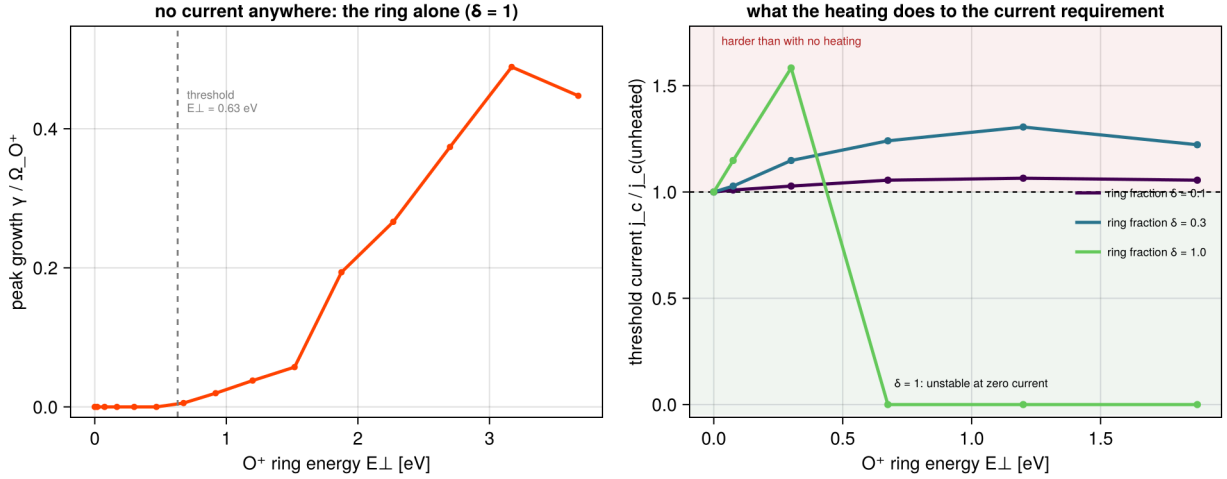


Figure 1: **Left:** currentless growth versus O^+ perpendicular energy ($\delta = 1$); the dashed line is the $\gamma = 10^{-3}\Omega$ threshold. **Right:** the current the same band needs, normalised to the unheated value. Every curve starts by moving into the red – heating makes the current-driven mode harder – and only $\delta = 1$ escapes, by crossing into its own currentless instability.

6 The moments do not carry the information

A gyro-ring at sharpness v_r/w has $\langle v_{\perp}^2 \rangle = w^2 + v_r^2$, hence an apparent anisotropy $T_{\perp}/T_{\parallel} = 1 + (v_r/w)^2$. Pair each ring with the bi-Maxwellian carrying **identical** $n, T_{\perp}, T_{\parallel}$:

$T_{\perp}/T_{\parallel}$	ring γ/Ω	ring u_c/v_e	bi-Max γ/Ω	bi-Max u_c/v_e
1.00	0	0.1055	0	0.1055
1.25	0	0.1211	0	0.1201
2.00	0	0.1670	0	0.1582
3.25	+0.0055	0	0	0.2197
5.00	+0.0379	0	0	0.2939
7.25	+0.1937	0	0	0.4053

Table 4: Same first two velocity moments, opposite answers. At $T_{\perp}/T_{\parallel} = 7.25$ the ring needs no current at all; the bi-Maxwellian needs four times the isotropic threshold.

Note the bi-Maxwellian trend: raising $T_{\perp}$ at fixed moments-of-record makes the current-driven mode **harder** to excite, monotonically, because perpendicular broadening is pure cyclotron damping when there is no positive slope to go with it. The ring reaches the opposite extreme. Anywhere between “no current needed” and “4× the isotropic threshold” is compatible with the same reported $(n, T_{\perp}, T_{\parallel})$.

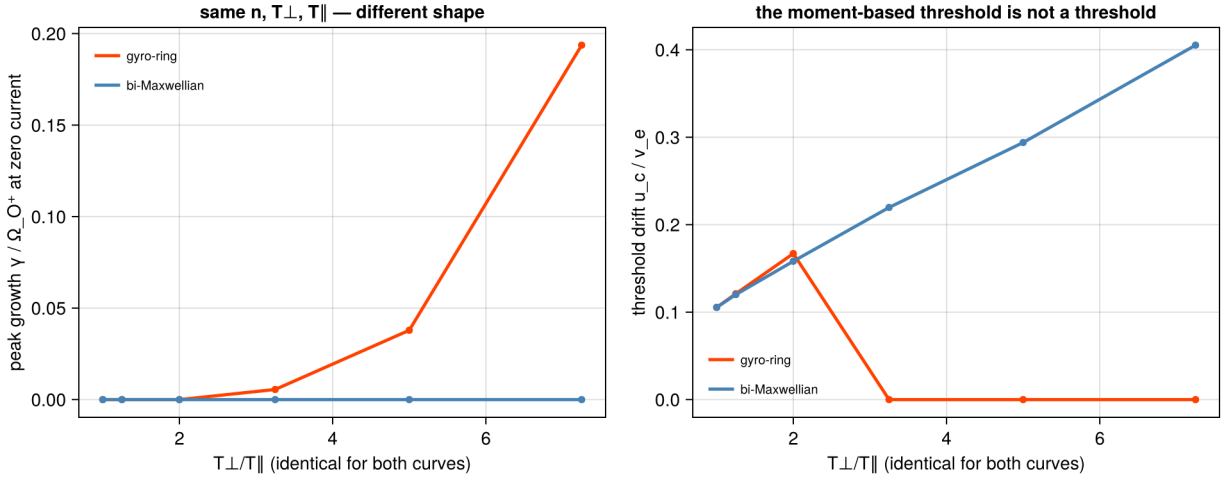


Figure 2: Ring versus moment-matched bi-Maxwellian. Every pair shares $n, T_{\perp}, T_{\parallel}$ exactly.

7 What this means

Transverse heating is not the explanation for sub-threshold EIC. We set out to test whether the co-observed conic removes the need for the shear mechanism. It does not: in the regime the topside actually presents — a suprathermal heated minority over a cold core — it makes the discrepancy worse. The shear explanation is not displaced by this alternative.

Where the ring channel does operate, the population is right. The requirement $\delta \gtrsim 0.6$ is a strong statement about where to look, and it is consistent with where ring-driven O^+ cyclotron harmonics are actually reported — the inner magnetosphere and plasma sheet boundary layer, where the energetic O^+ is not sitting on top of a dense cold ionospheric core. The topside ionosphere is the one place the mechanism is **suppressed**, and it is suppressed for a reason that is measurable: the cold core fraction.

The prediction that survives is a distribution measurement. Where the O^+ perpendicular distribution is a ring with positive inner slope, $E_{\perp} \gtrsim 0.6$ eV, **and** the heated component dominates the local O^+ , the ion-cyclotron band should be unstable irrespective of the current, with power hopping between harmonics as the ring sharpens. Where the heated component is a minority, EIC should require **more** current than the classical threshold, not less — a signed, falsifiable prediction.

Threshold curves drawn in moment space are not thresholds. Any stability criterion for ion outflow expressed in $(n, T_{\perp}, T_{\parallel})$ — as parameterisations of auroral heating and outflow generally are — is under-

determined by the range shown in §6. This is the same lesson the solar-wind anisotropy constraint teaches, arriving in the ionosphere.

A caution on self-consistency. The threshold is on ring **sharpness** v_r/w , not energy: what destabilises the band is the positive perpendicular slope, and a heating process that broadens the shell as fast as it displaces it never reaches threshold at any energy. The energies quoted assume the heated population keeps roughly the core thermal width. Whether real TAI distributions do is measurable and is the sharpest way to falsify this.

Scope. Local, homogeneous, electrostatic, collisionless, single ion species, linear. The companion note `experiments/shear-kinetic/theory.md` treats the inhomogeneous (sheared) problem and shows its drive lives entirely in the flow-advection term — so the two candidate mechanisms are physically distinct and do not overlap.

8 Reproducing

`01_gate_current.jl` → `02_ring_threshold.jl` → `03_joint_threshold.jl` → `04_moment_control.jl` → `05_make_figures.jl`. Shared model and drivers in `common.jl`; each script writes its own `out_*.tsv`.